

NanoGT Scoring Evidence Table

Working document for linking each NanoGT scoring rule to evidence, rationale, and limitations. This should be treated as a dissertation audit trail, not as a claim that every threshold is statistically calibrated.

Bottom Line

NanoGT currently uses an evidence-informed heuristic score. Some parameters are directly grounded in literature or clinical practice, especially AAV packaging capacity, vector seroprevalence, clinical-stage maturity, and known routes of administration. Other parts are expert-rule bins chosen to make the model interpretable across a small proof-of-concept cohort.

Use this wording in the dissertation:

Scoring dimensions and thresholds were selected from literature-supported biological and development constraints where available, supplemented by transparent heuristic bins where no validated quantitative weighting scheme exists for rare-disease gene-therapy precedent matching. Composite scores therefore represent relative precedent strength, not calibrated probabilities of clinical success.

Evidence Status Key

Status	Meaning
Direct evidence	The parameter itself is supported by literature or clinical/regulatory fact.
Evidence-informed heuristic	The biological direction is supported, but the exact score or cutoff is a modelling choice.
Heuristic	The rule is mainly an interpretive choice and needs sensitivity analysis or expert validation.
Needs audit	Plausible, but the current project needs a checked paper, page note, or stronger citation.

Scoring Weights

Dimension group	Current max score	Current interpretation	Evidence status	Rationale	Limitation
Packaging, tropism, protein class, pathway, mechanism/modality compatibility, immunogenicity, therapeutic window	2.0 each	Higher-weight determinants	Evidence-informed heuristic	These are major feasibility constraints: the gene must fit, vector must reach tissue, biology must be transferable, disease mechanism must suit the therapeutic modality, immune barriers matter, and timing affects trial feasibility.	The choice to give exactly double weight is not statistically derived.
Inheritance, approval, cross-correction, immune privilege, promoter availability, route feasibility	1.0 each	Lower-weight determinants	Heuristic	These modify precedent strength but are not treated as full feasibility gates in v0.1.	Some of these may deserve higher weight in specific diseases, especially route, cross-correction, and approval precedent.

Dimension group	Current max score	Current interpretation	Evidence status	Rationale	Limitation
Organelle targeting	1.0	Penalises diseases where standard nuclear AAV delivery cannot produce a functional protein at the correct subcellular compartment (e.g. mtDNA-encoded genes, nuclear-encoded mitochondrial matrix proteins)	Evidence-informed heuristic	Added in v2 after MT-ND4/LHON received a plausible-looking 6.8/10 without any flag that allotopic expression is an entirely different paradigm from standard nuclear gene addition. MUT/MMA similarly received no penalty for the unvalidated MTS import requirement. Convenient normalisation denominator. Organelle targeting added in v2 to penalise mitochondrial DNA-encoded genes and nuclear-encoded mitochondrial matrix proteins.	Only scores the two most common organelle-targeting failure modes (mtDNA and nuclear-encoded mitochondrial matrix). Peroxisomal and lysosomal membrane targeting are captured separately by the protein-class and cross-correction dimensions.
Raw total	21.0 (v2, +1.0 for organelle targeting)	Sum of all dimension maxima	Heuristic	Convenient normalisation denominator. Organelle targeting added in v2 to penalise mitochondrial DNA-encoded genes and nuclear-encoded mitochondrial matrix proteins.	Not a trained model; weights were not fitted to outcome data.
Confidence tiers	High ≥ 7.5 , medium ≥ 5.0 , low < 5.0	Interpretive score bands	Heuristic	Makes outputs easier to discuss and compare.	Needs sensitivity analysis; does not correspond to probability of success.

Dimension-Level Evidence Map

#	Dimension	Current rule	Evidence status	Evidence/ratios/sources	Dissertation- safe	Limitation/action
					wording	
1	Packaging fit	CDS > vector cargo gives hard fail. Otherwise utilisation <=30% = 2.0; 31-60% = 1.5; 61-85% = 1.0; 86-100% = 0.5.	Direct evidence for hard gate; heuristic for bins.	AAV practical packaging limit is supported by Grieger and Samulski 2005 and AAV reviews such as Naso et al. 2017 and Wang et al. 2024. Chamberlain et al. 2016 supports alternative strategies for oversized genes. Chamberlain et al. 2023 supports DMD microdys-trophin logic.	The hard gate reflects a physical single-vector packaging constraint. Sub-bins represent increasing engineering risk as cargo approaches capacity.	Current score uses coding sequence only, not full vector genome including promoter, polyA, ITRs, regulatory elements, and codon optimisation. Add sensitivity analysis around 70%, 85%, and full cassette size.

#	Dimension	Current rule	Evidence status	Evidence/ratios/sources	Dissertation- male	
					wording	Limitation/action
2	Tissue tropism	2.0 if vector tropism and precedent target both match disease tissue; 1.5 for target match or multi-tissue tropism overlap; 1.0 for single vector overlap; 0.3 for no overlap.	Evidence-informed heuristic.	Zincarelli et al. 2008 supports serotype-dependent tissue tropism. Foust et al. 2009 supports AAV9 CNS relevance. Walkey et al. 2025 provides a broad AAV tropism atlas in mouse. Clinical precedent entries in the catalog also support tissue-route pairings.	Tissue overlap is used as a practical proxy for vector transferability. Higher scores require both biological tropism and precedent-program targeting.	Tropism is species-, dose-, promoter-, route-, age-, and capsid-engineering dependent. Mouse atlas evidence does not directly translate to human efficacy. Consider separating natural tropism from route-specific clinical evidence.

#	Dimension	Current rule	Evidence status	Evidence sources	Dissertation- safe	
					wording	Limitation/action
3	Protein class	Lysosomal or secreted exact match = 2.0; intracellular or membrane match = 1.5; partial secreted/lysosomal component = 1.0; mismatch = 0.5.	Evidence-informed heuristic.	Gene therapy reviews and lysosomal storage disease prece-dents support differ-ences between secreted/cross-correctable proteins and cell-autonomous intracellu-lar proteins. UniProt supports protein lo-calisation annotations.	Protein class is used as a proxy for whether the prece-dent's mecha-nism transfers to the query disease.	Current categories are coarse. It should dis-tinguish secreted lysosomal enzymes, non-secreted lysosomal mem-brane proteins, cytosolic proteins, nuclear proteins, mitochon-drial proteins, and structural mem-brane proteins.

#	Dimension	Current rule	Evidence status	Evidence/ratios/sources	Dissertation-safe	
					wording	Limitation/action
4	Mechanism/compatibility	Compatible gene addition = 2.0; conditional = 1.5; uncertain = 1.0; incompatible = 0.0.	Evidence-informed heuristic with source-linked disease facts.	data/disease stores the dis-ease/gene mechanism category, gene-addition fit, evidence summary, and evidence URL. General gene therapy reviews support gene addition as most straight-forward for loss-of-function disease, while dominant-negative, toxic gain-of-function, mitochondrial, repeat-expansion, and RNA/splicing disorders may need other modalities.	The mechanisms work now checks whether the disease mechanism supports ordinary gene addition before treating vector precedent as transferable.	Thev curated mechanism table is only as good as its citations. Conditional labels still require expert review and should not be converted into clinical recommendations.

#	Dimension	Current rule	Evidence status	Evidence/ratios/sources	Dissertation- safe	
					wording	Limitation/action
5	Inheritance compatibility	Exact AR/XL match = 1.0; shared loss-of-function = 0.7; mismatch/dominant/mitochondrial = 0.3.	Evidence-informed heuristic.	General gene therapy reviews, including High and Roncarolo 2019 and Kirschner and Cathomen 2020, support gene addition as most straightforward for loss-of-function monogenic disease.	Inheritance remains useful context, but it is no longer treated as sufficient evidence for mechanism.	Exact scores are heuristic. Inheritance can mislead: dominant disease can be haploinsufficiency or dominant-negative; X-linked disease can still require tissue-specific interpretation.

#	Dimension	Current rule	Evidence status	Evidence sources	Dissertation- safe	Limitation/action
					wording	
6	Pathway similarity	Exact or grouped pathway match = 2.0; related pathway = 1.5; different pathway = 0.5; unknown = 1.0.	Evidence-informed heuristic.	Pathway labels are curated from disease biology, UniProt/GO/HPD terms, and clinical precedent groupings. Disease-similarity literature in the Zotero export, including Wei et al. 2024 and Lagorce et al. 2024, supports phenotype/pathway similarity as a computational strategy in rare disease analysis.	Shared pathway increases precedent relevance because of assay, biomarkers, natural history, and regulatory analogies may transfer better.	Groupings are hand-coded and not learned from ontology distances. Consider using HPO/ORDO/GO semantic similarity rather than manual pathway buckets.

#	Dimension	Current rule	Evidence status	Evidence/rational sources	Dissertation- in file	
					wording	Limitation/action
7	Regulatory approval weight	Approved = 1.0; phase3 = 0.8; withdrawn/phase2-3 = 0.7; phase2 = 0.6; phase1-2 = 0.5; phase1 = 0.4; unknown = 0.3.	Evidence-informed heuristic.	Clinical-stage maturity is a reasonable proxy for precedent strength. Lomash et al. 2023 and Lomash et al. 2025 support the importance of regulatory planning, ODD/RPDD, TPP, and FDA interaction in rare-disease AAV development. FDA/EMA guidance in the bibliography supports regulatory maturity as a development constraint.	Later-stage or approved programs are stronger development precedents than early-stage programs.	Numerical spacing is heuristic. Withdrawn status is ambiguous: it may still prove technical feasibility but can signal commercial, safety, or efficacy limitations.

#	Dimension	Current rule	Evidence status	Evidence/ratios/sources	Dissertation- safe	
					wording	Limitation/action
8	Vector immunogenicity	Seroprevalence 0% = 2.0; 10-19% = 1.5; 20-39% = 1.0; ≥40% = 0.5.	Direct evidence for seroprevalence values; heuristic for bins.	Boutin et al. 2010, Calcedo and Wilson 2013, Mingozzi and High 2013, Verdera et al. 2020, and Wang et al. 2024 support AAV humoral immunity as a barrier. Current code values: AAV5 9%, AAVrh10 10%, AAV2/6 17%, AAV1 20%, AAV9 22%, AAV8 30%, AAV2 55%, LV 2%.	Pre-existing anti-capsid immunity is treated as a vector-level eligibility and development risk. Lower estimated seroprevalence receives a higher score.	Seroprevalence varies by geography, age, assay, neutralisation threshold, population, and cross-reactivity. Does not model dose, complement, T-cell responses, steroid regimen, anti-transgene immunity, or redosing.

#	Dimension	Current rule	Evidence status	Evidence/ratios/sources	Dissertation- male	Limitation/action
					wording	
9	Therapeutic window	Adult/chronic = 2.0; progressive childhood = 1.5; early childhood = 1.2; neonatal = 0.8; congenital/rapidly fatal = 0.5.	Evidence-informed heuristic.	Kirschner and Cathomen 2020 supports timing importance, including stronger effects when SMA gene therapy is administered before symptoms. Natural-history logic and HPO terms support using disease timing as a trial-feasibility constraint.	Diseases needing very early treatment are not impossible, but clinical translation is harder because irreversible damage may occur before dosing.	Exact bins are heuristic and disease-name keyword matching is crude. Should use disease-specific natural history, newborn screening availability, and age at irreversible pathology when possible.

#	Dimension	Current rule	Evidence status	Evidence/ratios/sources	Dissertation- male	
					wording	Limitation/action
10	Cross-correction	Secreted or secreted lysosomal = 1.0; lysosomal non-secreted = 0.8; intracellular/membrane = 0.2.	Direct biological rationale; heuristic numeric bins.	Lysosomal storage disease and ex vivo HSC literature, including Biffi et al. 2013, supports cross-correction as a powerful mechanism for secreted/lysosomal enzymes. General gene therapy biology supports lower rescue potential for intracellular/membrane proteins.	Cross-correctable proteins can benefit untransduced cells, reducing the required fraction of target-cell transduction	Current code is too coarse. Some lysosomal membrane proteins, secreted proteins with poor uptake, and structural membrane proteins need separate treatment.
11	Immune privilege	Retina = 1.0; CNS = 0.9; liver = 0.8; muscle/heart = 0.6; kidney = 0.5; hematopoietic = 0.3.	Evidence-informed heuristic.	Retina, CNS, and liver immune environments are widely discussed in gene therapy reviews and clinical precedent. Wang et al. 2024 reviews immune responses and safety concerns in AAV gene therapy.	Target tissues differ in immune surveillance and durability risk, so tissue context modifies precedent strength.	Needs stronger tissue-specific citations. Current values do not model route, dose, capsid, promoter, inflammation, immunosuppression, or disease-specific immune activation.

#	Dimension	Current rule	Evidence status	Evidence/ratios/sources	Dissertation- safe	
					wording	Limitation/action
12	Promoter availability	Liver/retina = 1.0; CNS/muscle = 0.8; hematopoietic = 0.7; heart = 0.6; kidney = 0.4.	Evidence-informed heuristic.	Clinical programs in the catalog support pro-moter/tissue are precedent: liver AAV programs, Lux-turna/retinal programs, Elevidys/MHCK7-style muscle expression, lentiviral HSC programs. PaVe-GT papers support the importance of vector design and product planning.	Tissues with multiple clinically used promoters are treated as easier to engineer safely and effectively.	Needs explicit citations per promoter. Current score does not distinguish cell-type specificity, promoter size, expression strength, disease cell type, or silencing risk.

#	Dimension	Current rule	Evidence status	Evidence sources	Dissertation- safe	
					wording	Limitation/action
13	Route of administration feasibility	Liver = 1.0; muscle = 0.9; hematopoietic = 0.9; retina = 0.8; CNS = 0.7; heart = 0.6; kidney = 0.4.	Evidence-informed heuristic.	Approved/late stage program routes support this: IV liver AAV for hemophilia/metabolic programs, subretinal/intravitreal ocular programs, ex vivo HSC lentiviral programs, intrathecal/ICV CNS programs, systemic/IM muscle programs. Wang et al. 2024 and PaVe-GT papers support route and development planning as major constraints.	Established clinical delivery routes increase precedent transferability	Values are heuristic. Route feasibility depends on target cell type, required biodistribution, procedural risk, age, dose, immune toxicity, and whether partial tissue correction is sufficient.

#	Dimension	Current rule	Evidence status	Evidence/ratio sources	Dissertation- male	
					wording	Limitation/action
14	Organelle targeting feasibility	mtDNA-encoded gene (MT-* symbol or mitochondrial inheritance) = 0.0; nuclear-encoded mitochondrial matrix protein (UniProt annotation + MTS signal) = 0.5; peroxisomal targeting required = 0.7; no special organelle targeting required = 1.0.	Evidence-informed heuristic with source-linked disease facts.	Standard nuclear AAV delivers a transgene to the nucleus where it is transcribed and translated by cytoplasmic ribosomes. For most disease genes this is sufficient. mtDNA-encoded genes require allotopic expression: cytoplasmic recoding plus an artificial MTS (real-world precedent: GS010/Lumevoq for MT-ND4 LHON; EMA MAA withdrawn April 2023 after primary endpoint not met). Nuclear-encoded mitochondrial matrix proteins (e.g. MUT in MMA) require the therapeutic construct to include an intact N-	Makes a fundamental delivery mismatch numerically visible in the composite score.	Only models mtDNA and nuclear-encoded mitochondrial matrix failure modes explicitly. Peroxisomal (PTS1/PTS2) and lysosomal membrane protein issues are partially captured by other dimensions. Does not model partial allotopic success, heteroplasmy level, or tissue-specific mtDNA copy number.

#	Dimension	Current rule	Evidence status	Evidence/ratios/sources	Dissertation- safe wording	Limitation/action
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Source Mapping From Zotero Export

The exported Zotero file is Reading Literature/Reading Literature.bib. The PDFs have also been extracted to searchable text files in Reading Literature/text/ using extract_pdfs.py.

Zotero key	Useful for	Evidence use
chamberlain_microdystrophin_2023	DMD microdystrophin, oversized native DMD, accelerated approval/surrogate endpoint	Supports the DMD micro/mini-transgene exception and the statement that full-length dystrophin exceeds single AAV capacity.
brooks_platform_2020	PaVe-GT platform development	Supports platform-vector rationale and standardised rare-disease AAV development framing.
lomash_successfully_2023	Orphan drug and rare pediatric disease designations	Supports regulatory-development dimension and rare disease program constraints.
lomash_adeno-associated_2025	AAV development planning, TPP, FDA INTERACT	Supports regulatory planning, early development, and product design constraints.
wang_adeno-associated_2024	AAV review, immune responses, clinical use, manufacturing and safety concerns	Supports immunogenicity, vector safety, AAV clinical breadth, and delivery-vector limitations.
walkey_comprehensive_2025	AAV tropism atlas	Supports tropism as a real vector property, while requiring caution about mouse-to-human translation.
kirschner_gene_2020	Monogenic gene therapy overview	Supports gene addition framing, timing importance, and in vivo/ex vivo distinction.
rath_representation_2012	Orphanet methodology	Supports use of Orphanet as rare disease source.
lagorce_phenotypic_2024	HPO/ORDO phenotypic similarity	Supports using phenotype similarity methods in rare disease computational pipelines.
wei_dismvc_2024	Disease similarity modelling	Supports computational disease-similarity framing, but not gene therapy-specific scoring.
georgeson_bionitio_2019	Bioinformatics command-line tool practices	Supports CLI, packaging, documentation, tests, reproducibility framing.
leprevost_best_2014	Bioinformatics software best practices	Supports software engineering/reproducibility methods.

Page-Checked Local Text Anchors

These local text anchors come from the extracted Zotero PDFs. Use them to trace the strongest scoring claims back to exact readable passages.

Claim	Local extracted-text anchor	How it supports the scoring table
Full-length dystrophin exceeds single-AAV packaging capacity; microdystrophin is the relevant engineered strategy.	Reading Literature/text/chamberlain-et-al-2023-microdystrophin-expression-as-a-surrogate-endpoint-for-duchenne-muscular-dystrophy-clinical-trial.txt, lines 24-29 and 214-218.	Supports the DMD native-gene hard fail and the DMD_micro exception added to NanoGT.
Microdystrophin constructs are shortened functional dystrophin genes intended to retain key function.	Same Chamberlain text file, lines 95-97, 218-226, and 253-255.	Supports treating SRP-9001/DMD_micro as an engineered-cargo precedent rather than as ordinary full-length gene replacement.
AAV therapy faces immune barriers including pre-existing immunity and neutralising antibodies.	Reading Literature/text/wang-et-al-2024-adenovirus-as-a-delivery-vector-for-gene-therapy-of-human-diseases.txt, lines 2200-2208 and 2297-2308.	Supports including vector immunogenicity as a scoring dimension.
AAV antibody interactions can block target-cell entry; complement activation is observed with high-dose rAAV.	Same Wang text file, lines 2329-2337 and 2377-2419.	Supports the limitation that immunogenicity is broader than simple seroprevalence and includes complement/toxicity risks.
AAV tropism is tissue- and cell-type dependent and varies across species.	Reading Literature/text/walkey-et-al-2025-a-comprehensive-atlas-of-aav-tropism-in-the-mouse.txt, lines 145-152 and 888-900.	Supports using tropism while explicitly warning that mouse tropism is not directly human efficacy.
AAV tropism studies measure biodistribution and functional transduction across tissues.	Same Walkey text file, lines 180-189, 324-326, and 350-362.	Supports treating vector/tissue overlap as evidence-informed rather than arbitrary.
Target Product Profile and early FDA interaction guide AAV gene therapy development.	Reading Literature/text/lomash-et-al-2025-adenovirus-gene-therapy-development-early-planning-and-regulatory-considerations-to-adv.txt, lines 21-34 and 187-208.	Supports the regulatory maturity/planning dimension as a development constraint.

Claim	Local extracted-text anchor	How it supports the scoring table
FDA INTERACT meetings can address preclinical models, safety, biodistribution, CMC, immunogenicity, and clinical design.	Same Lomash text file, lines 217-231 and 424-436.	Supports interpreting approval/regulatory stage as more than a label: it reflects accumulated development evidence and regulator interaction.

Sources Already In paper/references.md

These are not all in the Zotero export, but they are already in the project bibliography and map directly to scoring claims.

Reference number	Source	Scoring relevance
5	Russell et al. 2017	Luxturna, retinal route, AAV2 precedent.
6-7	Mendell et al. 2017; Day et al. 2021	Zolgensma/SMA, AAV9, timing of early treatment.
8	Pipe et al. 2023	Hemgenix, liver AAV5 precedent.
9	Pasi et al. 2020	Roctavian/large AAV cargo hepatic precedent.
10	Mendell et al. 2020	DMD microdystrophin clinical precedent.
13	Zincarelli et al. 2008	AAV serotype tropism.
14	Foust et al. 2009	AAV9 CNS relevance.
15	Grieger and Samulski 2005	AAV packaging capacity.
16	Naso et al. 2017	AAV vector review.
17	Chamberlain et al. 2016	Oversized transgenes and engineering strategies.
22-25	Boutin et al. 2010; Calcedo and Wilson 2013; Mingozi and High 2013; Verdera et al. 2020	AAV immunogenicity and seroprevalence.
26	Biffi et al. 2013	Lentiviral HSC and lysosomal/leukodystrophy cross-correction precedent.
34-36	FDA/EMA guidance	Regulatory maturity, CMC, and clinical/non-clinical development framing.

Methods Text To Add

Suggested paragraph for paper/methods.md, after the sentence introducing the 13 scoring dimensions:

The scoring scheme was not trained against clinical outcome data. Instead, thresholds were selected using a transparent evidence-informed heuristic approach. Where quantitative constraints were available, such as approximate vector cargo capacity and published AAV seroprevalence estimates, these were used directly. Where no validated quantitative weighting scheme exists, such as assigning relative importance to pathway similarity, promoter availability, or route feasibility, rule-based bins were

chosen to preserve interpretability and allow sensitivity analysis. Scores should therefore be interpreted as relative precedent-strength estimates rather than probabilities of clinical efficacy.

Immediate Gaps To Fix Before Submission

1. Add page-checked citations for promoter availability by tissue.
2. Add page-checked citations for immune privilege by tissue.
3. Add page-checked citations for route feasibility categories.
4. Source-audit every row in `data/disease_mechanisms.csv` beyond the currently checked cohort-level links.
5. Run a sensitivity analysis with alternate weights, for example equal weights for all dimensions and a stricter route/tropism weighting.
6. Move exact threshold values into a documented configuration or methods table if the project continues beyond the dissertation proof-of-concept.